DAY 31 | HANDS-ON ACTIVITIES

# LLMOps Intro and FDE Skills 4: Problem Statements

Four technical problems and three people-skills activities.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

**How to work:** read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

## Problem 1: Read the Incident Logs

**Level:** Warm-up | **Concepts:** log analysis

Using the notebook's simulated week of LLM logs, find when errors spiked, which error code dominated, and when cost per request jumped.

### Sample run / expected result

```
Errors peak 10:00 to 11:00 on 2026-12-02, mostly 429
Cost per request triples from 2026-12-04
```

### Hints

- Group by hour and status; plot tokens per request by day.

## Problem 2: Backoff with Jitter

**Level:** Easy | **Concepts:** retries

Wrap the flaky provider in a retry function with exponential backoff and jitter. Measure success rate with 0, 2 and 4 retries.

### Sample run / expected result

```
retries=0: 70% | retries=2: 96% | retries=4: 99%
```

### Hints

- Retry only transient errors, never 400 Bad Request.

## Problem 3: Fallback and Circuit Breaker

**Level:** Medium | **Concepts:** resilience patterns

Add a fallback model and a circuit breaker that opens after 3 consecutive failures for 5 seconds. Show the user always gets a response.

### Sample run / expected result

```
breaker OPEN -> served by fallback
after cool-down -> HALF-OPEN -> CLOSED
```

### Hints

- Track consecutive failures and the time the breaker opened.

## Problem 4: Classify Severity

**Level:** Easy | **Concepts:** incident severity

Assign SEV1 to SEV4 to six incidents (data leak, total outage, slow at peak, wrong answers for one topic, typo, cost spike) and justify each.

### Sample run / expected result

```
Data leak: SEV1 (data at risk)
Slow at peak: SEV3 (degraded, usable)
```

### Hints

- Severity is about impact, not about how hard it is to fix.

## Problem 5: Write the Escalation

**Level:** Writing, individual, 15 minutes | **Concepts:** escalation structure

Write the escalation for today's simulated incident using what, impact, evidence, actions so far, ask and next update.

### Sample run / expected result

SEV2: Support assistant failing for 40% of requests since 10:05 IST due to provider rate limits.
Impact: about 1,200 customers saw errors.
Actions: enabled backoff and fallback; errors now 8%.
Ask: approve a rate-limit increase request today.
Next update: 11:30 IST.

### Hints

- Numbers and times make an escalation credible.

## Problem 6: The Frustrated Customer

**Level:** Role play, pairs, 2 rounds of 8 minutes | **Concepts:** empathy, self-regulation

The customer's head of operations calls during the incident: "This is the third time this month. Your AI is making us look bad." Respond as the FDE.

### Sample run / expected result

Acknowledge: "You are right to be frustrated; this affected your customers."
Facts: what happened and what is already fixed
Plan: what changes to prevent a repeat
Next update time, and keep it

### Hints

- Do not defend or blame. Lower your pace.
- Write down their exact concern.

## Problem 7: Blameless Review

**Level:** Group, 20 minutes | **Concepts:** post-incident review

Write a one-page blameless review: timeline, root causes, what went well, and three actions with owners and dates.

### Sample run / expected result

Root cause: no backoff, single model, no alert on 429 rate
Action: add alert at 5% errors (owner: Ravi, 2026-12-09)

### Hints

- Ask "how did the system allow this?" not "who did this?"

*Hou-ren-sou (報連相): report, inform, consult*